

Module 1 Assignment — Regression Diagnostics with R

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ALY6015: Intermediate Analytics

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September, 20th 2025

Regression Diagnostics and Model Selection for Ames Housing Data

09/20/2025

Introduction:

The ability to accurately predict the sale price of residential properties is a cornerstone of real estate market analysis, benefiting sellers, buyers, and financial institutions alike. The Ames Housing dataset, containing 2,930 observations and 82 variables describing properties sold in Ames, Iowa from 2006 to 2010, provides a rich opportunity for predictive modelling. The primary goal of this analysis is to build, evaluate, and refine a multiple linear regression model to predict property sale prices.

This report documents the systematic process of this endeavour, beginning with exploratory data analysis (EDA) to understand the data's structure and relationships. We will then construct an initial regression model and perform a suite of diagnostic tests to assess its validity and identify potential issues such as multicollinearity, non-linearity, and the influence of outliers. Based on these diagnostics, we will implement corrective measures to improve the model's performance and reliability. Finally, we will employ an automated feature selection technique, all subsets regression, to identify an optimal model. This report concludes by comparing the manually refined model with the algorithmically selected one, offering a final recommendation for the preferred model based on statistical rigor and interpretability.

Analysis:

The analysis follows a structured, multi-step approach from data preparation to final model comparison. All R code used for this analysis is provided in the Appendix.

Exploratory Data Analysis & Data Preparation:

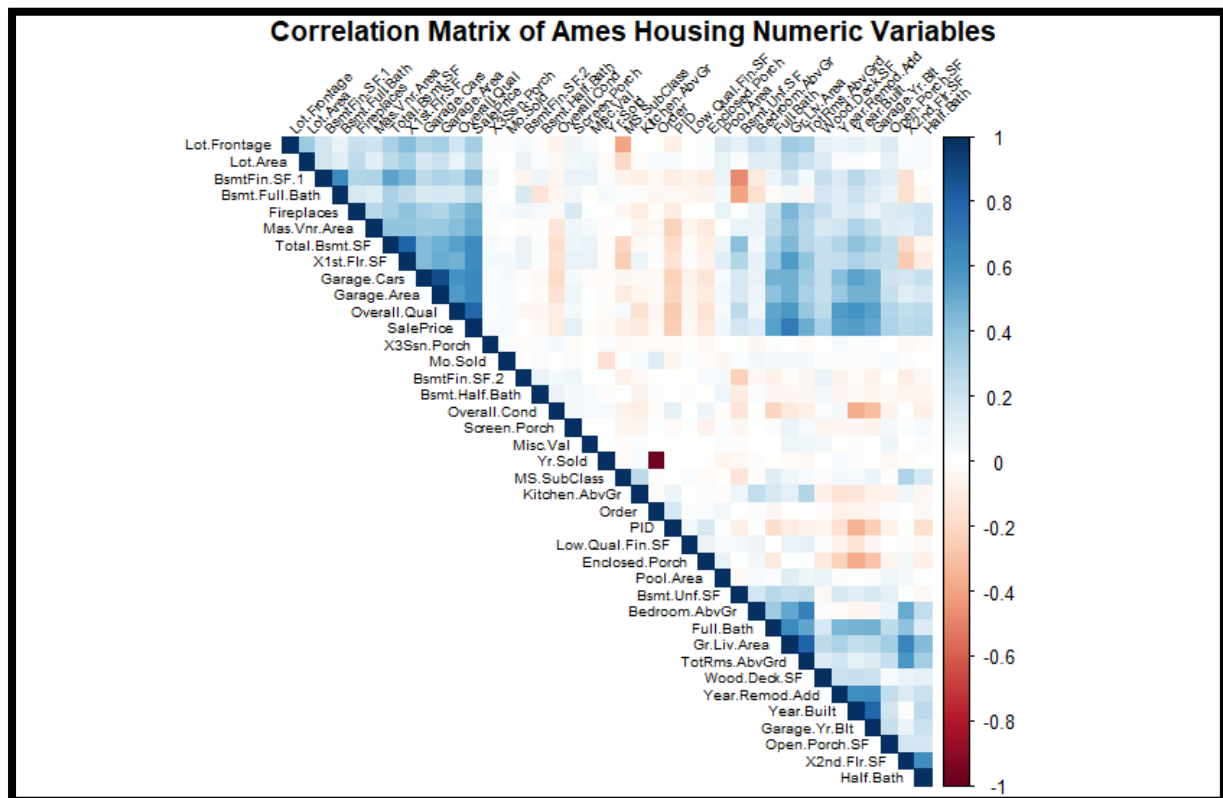
The analysis began by loading the Ames Housing dataset into R. An initial exploration revealed a mix of data types, including nominal, ordinal, discrete, and continuous variables. The target variable, SalePrice, is a continuous variable with a mean of \$180,796 and a median of \$160,000, indicating a right-skewed distribution.

A critical preprocessing step was handling missing data. For this analysis, we focused on the numeric variables. Missing values within these columns were imputed using their respective mean values. This approach is straightforward and ensures that we can proceed with correlation and regression analyses without losing a significant number of observations. For instance, the Lot.Frontage variable had missing values that were filled with its mean of 69.05.

Correlation Analysis:

To understand the linear relationships between numeric variables, particularly with SalePrice, a correlation matrix was computed. The corrrplot package was used to visualize this matrix, providing an intuitive overview of the strength and direction of these relationships.

Figure 1: Correlation Matrix of Numeric Variables



Interpretation: The plot uses colour intensity and hue to represent the correlation coefficient. Deep blue indicates a strong positive correlation (close to +1), while deep red signifies a strong negative correlation (close to -1). As expected, SalePrice showed the strongest positive correlations with variables related to property size and quality. The top three were:

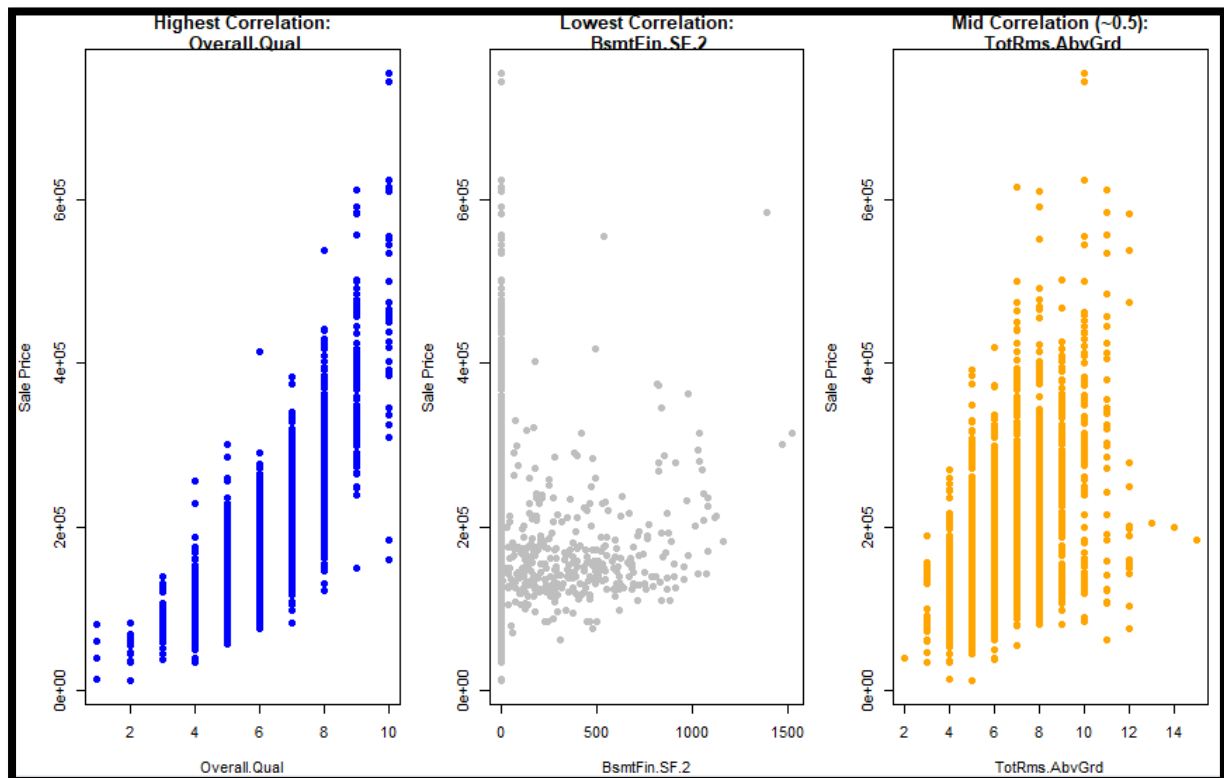
1. **Overall.Qual** (Overall Quality): $r = 0.799$
2. **Gr.Liv.Area** (Above Grade Living Area): $r = 0.706$
3. **Garage.Cars** (Size of garage in car capacity): $r = 0.647$

This confirms that higher quality homes with more living space tend to command higher prices.

Investigating Linear Relationships with Scatter Plots:

To visually inspect the relationships identified in the correlation matrix, three scatter plots were generated, comparing SalePrice against variables with high, low, and medium correlation.

Figure 2: Scatter Plots of SalePrice vs. Selected Predictors



1. **Highest Correlation (Overall.Qual):** The plot on the left shows a clear positive, though somewhat discrete, linear trend. As the overall quality rating increases, the sale price also systematically increases. There is more variance in SalePrice at higher quality levels.
2. **Lowest Correlation (BsmFin.SF.2):** The middle plot shows no discernible pattern. The points are scattered randomly, forming a cloud-like shape. This visually confirms the very weak linear relationship ($r = -0.006$) and suggests this variable would be a poor predictor of SalePrice on its own.
3. **Mid Correlation (Year.Remod.Add):** The plot on the right (correlation of $r = 0.505$) shows a moderately positive trend. Newer or more recently remodelled homes tend to have higher sale prices, but the relationship is less tight and contains more scatter than the Overall.Qual plot.

Initial Multiple Regression Model:

Based on the strong correlations and general real estate knowledge, an initial multiple regression model was fitted using three continuous variables: Overall.Qual, Gr.Liv.Area, and Total.Bsmt.SF.

The resulting model summary was highly significant ($p < 2.2e-16$), with an Adjusted R-squared value of 0.757, indicating that approximately 75.7% of the variance in SalePrice can be explained by these three predictors.

The model equation is: **SalePrice = -99259.19 + 23528.06 * Overall.Qual + 55.48 * Gr.Liv.Area + 57.06 * Total.Bsmt.SF**

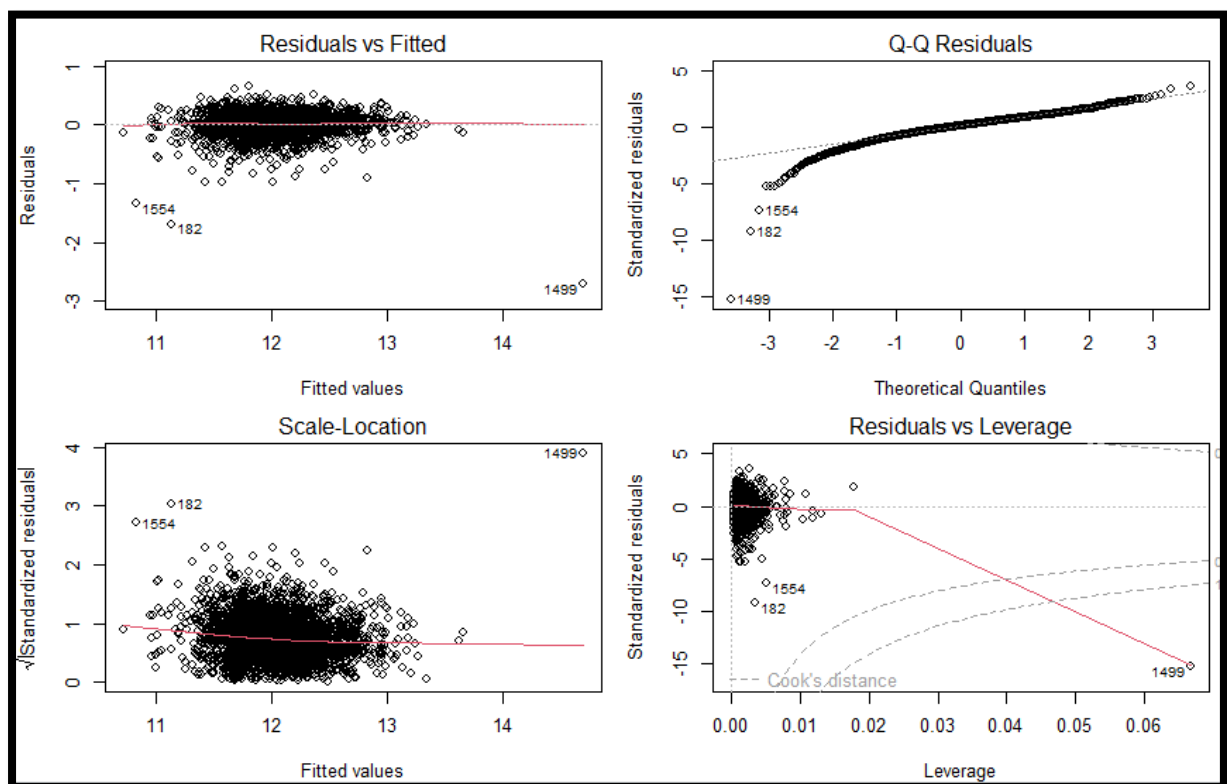
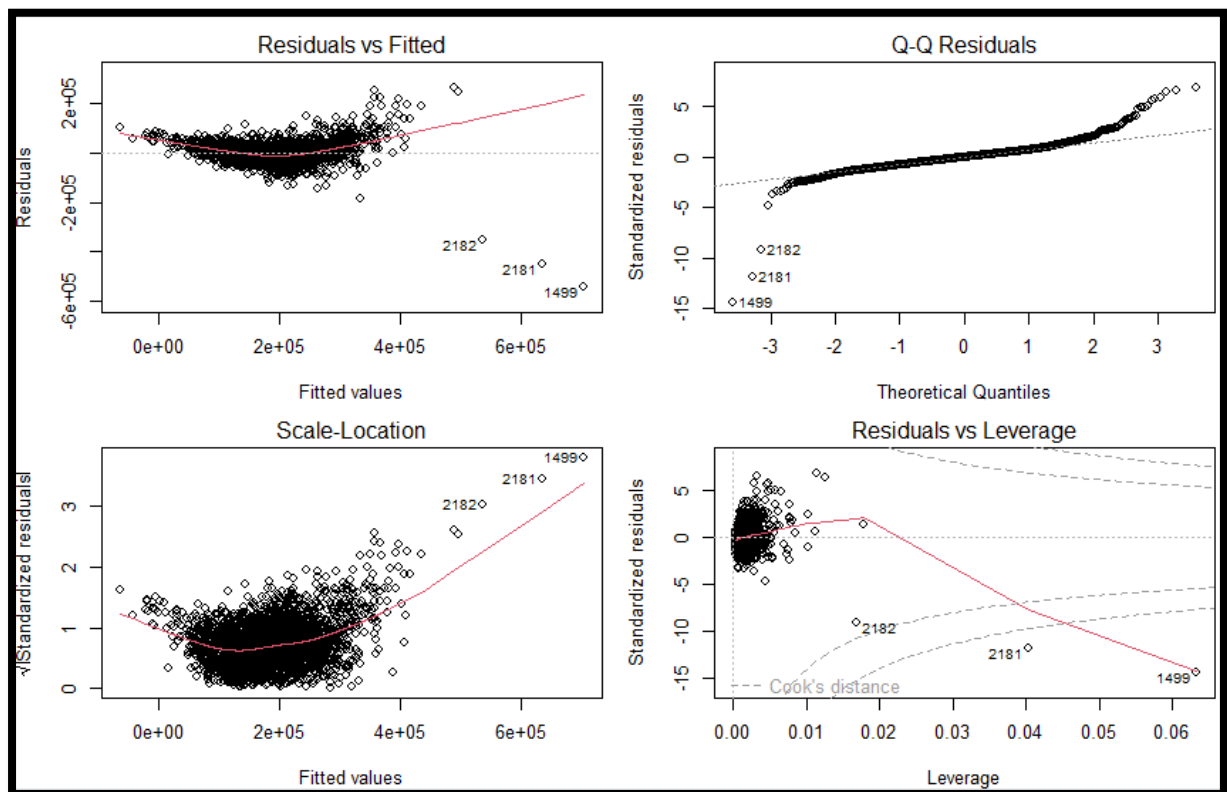
Coefficient Interpretation:

- **Intercept (-99,259.19):** This is the theoretical price of a house with zero for all predictor values. In this context, it is not practically interpretable.
- **Overall.Qual (23,528.06):** For each one-point increase in the overall quality rating, the predicted sale price increases by approximately \$23,528, holding living area and basement size constant.
- **Gr.Liv.Area (55.48):** For each additional square foot of above-grade living area, the predicted sale price increases by about \$55.48, holding quality and basement size constant.
- **Total.Bsmt.SF (57.06):** For each additional square foot of basement area, the predicted sale price increases by about \$57.06, holding quality and living area constant.

Diagnostic Analysis of Initial Model:

Standard regression diagnostic plots were generated to check the assumptions of linear regression.

Figure 3: Diagnostic Plots for the Initial Model



1. **Residuals vs. Fitted:** This plot showed a clear pattern (a curve) and non-constant variance (heteroscedasticity), where the spread of residuals increases as the fitted value increases. This violates the assumption of homoscedasticity.
2. **Normal Q-Q:** The residuals deviated significantly from the diagonal line, especially at the tails, indicating that they are not normally distributed.
3. **Scale-Location:** This plot confirmed the issue of heteroscedasticity, showing an upward trend.
4. **Residuals vs. Leverage:** This plot highlighted several influential data points, notably observations 2181 and 2182, which had high leverage and large residuals.

Multicollinearity: A check using the Variance Inflation Factor (VIF) revealed scores of 1.76 for Overall.Qual, 1.54 for Gr.Liv.Area, and 1.34 for Total.Bsmt.SF. Since all values are well below the common threshold of 5 or 10, multicollinearity is not an issue in this model.

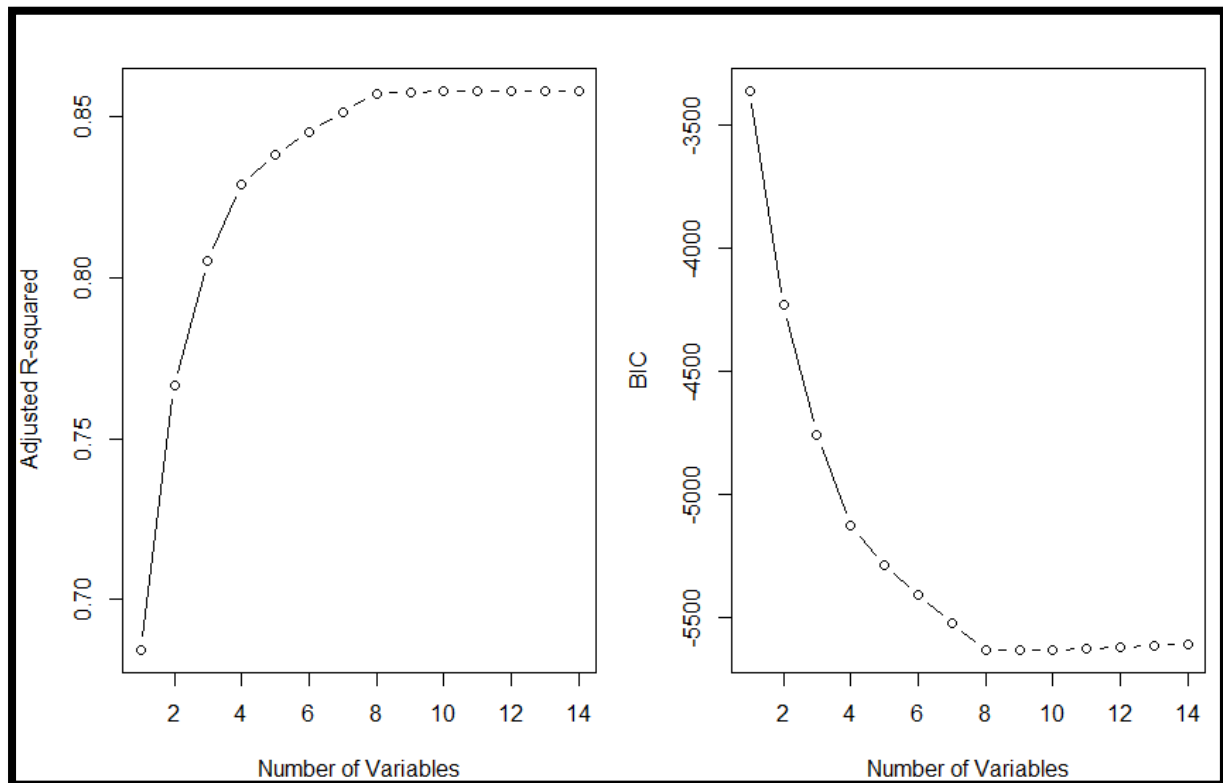
Outliers: The diagnostic plots identified observations 2181 and 2182 as significant outliers. These properties have exceptionally large living areas (Gr.Liv.Area > 5000 sq ft) but relatively low sale prices for their size, making them highly influential on the regression line. It was decided to remove these observations as they represent atypical sales that could distort a model intended for general use.

Model Improvement:

To address the issues of non-normality and heteroscedasticity, a logarithmic transformation was applied to the SalePrice variable. This is a common technique for handling right-skewed financial data. The two extreme outliers (2181, 2182) were also removed.

A new model was fitted: $\log(\text{SalePrice}) \sim \text{Overall.Qual} + \text{Gr.Liv.Area} + \text{Total.Bsmt.SF}$.

The improved model showed a significant increase in Adjusted R-squared to 0.846. The new diagnostic plots were much better, with the Residuals vs. Fitted plot showing less of a pattern and the Normal Q-Q plot indicating that the residuals are much closer to a normal distribution. This confirms that the transformation and outlier removal were effective.



Best Subsets Regression:

To explore if a better combination of predictors exists, the all subsets regression method was employed using the `regsubsets` function from the `leaps` package. This method fits models with all possible combinations of predictors and ranks them based on criteria like Adjusted R-squared or BIC (Bayesian Information Criterion). Using the top 14 predictors correlated with `SalePrice`, the analysis suggested that the best model, based on maximizing Adjusted R-squared, contained 11 variables.

This "best" model was then fitted: $\log(\text{SalePrice}) \sim \text{Overall.Qual} + \text{Gr.Liv.Area} + \text{Total.Bsmt.SF} + \text{Year.Built} + \text{BsmtFin.SF.1} + \text{Year.Remod.Add} + \text{Garage.Cars} + \text{X1st.Flr.SF} + \text{Fireplaces} + \text{Full.Bath} + \text{TotRms.AbvGrd}$

This model achieved an Adjusted R-squared of **0.890**, a substantial improvement over the initial model and a notable improvement over the manually refined 3-variable model.

Final Model Comparison:

- **Improved Manual Model (3 variables, log-transformed):**
 - Adjusted R-squared: 0.846
 - Variables: Overall.Qual, Gr.Liv.Area, Total.Bsmt.SF

- Interpretation: Simple, easy to interpret, and explains a large portion of the variance.
- **'Best' Subsets Model (11 variables, log-transformed):**
 - Adjusted R-squared: 0.890
 - Variables: Includes the original three plus others like Year.Built, Garage.Cars, etc.
 - Interpretation: More complex, but provides higher predictive accuracy by incorporating more housing features.

Preferred Model: For the purpose of achieving the highest predictive accuracy, the **'Best' Subsets Model with 11 variables is preferred**. While it is more complex, the 4.4% increase in explained variance (from 84.6% to 89.0%) is significant. This model provides a more nuanced understanding of the factors driving sale prices in Ames. If simplicity and ease of interpretation were the primary goals, the 3-variable model would be a very strong and defensible choice. However, in a predictive context, the model with higher accuracy is generally superior.

Conclusion:

This analysis successfully developed and refined a multiple linear regression model to predict housing prices in Ames, Iowa. The initial model, while statistically significant, suffered from violations of regression assumptions, specifically non-normal residuals and heteroscedasticity. Through diagnostic analysis, these issues were identified and corrected by applying a log transformation to the SalePrice variable and removing two extreme outliers.

The improved 3-variable model demonstrated strong performance, explaining 84.6% of the variance in the log of sale prices. However, an automated feature selection process using all subsets regression identified a more powerful 11-variable model that increased the explained variance to 89.0%. This final model confirms that house prices are driven by a combination of factors including overall quality, size (living area, basement, garage), age and remodeling status, and key features like fireplaces and bathrooms.

The key takeaway is that regression modeling is an iterative process. A statistically significant model is not necessarily a valid one. A thorough diagnostic investigation is crucial for building robust and reliable models. By systematically identifying and correcting issues, we were able to significantly enhance the model's performance and confidence in its predictive capabilities.

References:

De Cock, D. (2011). Ames, Iowa: Alternative to the Boston Housing Data Set. *Journal of Statistics Education*, 19(3).

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). *An Introduction to Statistical Learning: with Applications in R*. Springer.

Kutner, M. H., Nachtsheim, C. J., Neter, J., & Li, W. (2005). *Applied linear statistical models*. McGraw-Hill Irwin.

Appendix:

```
if (!require("corrplot")) install.packages("corrplot")
if (!require("ggplot2")) install.packages("ggplot2")
if (!require("leaps")) install.packages("leaps")
if (!require("car")) install.packages("car")

library(corrplot)
library(ggplot2)
library(leaps)
library(car)

tryCatch({
  ames_data <- read.csv("AmesHousing.csv")
  print("Ames Housing dataset loaded successfully.")
}, error = function(e) {
  print("Error loading AmesHousing.csv. Please ensure the file is in your working
directory.")
  stop(e)
})

# (EDA)

cat("\n--- Exploratory Data Analysis ---\n")
# Display the structure of the dataset
cat("\nStructure of the dataset:\n")
str(ames_data, list.len = 10)

# Display summary statistics for key variables
cat("\nSummary statistics for SalePrice:\n")
summary(ames_data$SalePrice)

cat("\nSummary statistics for some predictor variables:\n")
summary(ames_data[, c("Gr.Liv.Area", "Overall.Qual", "Total.Bsmt.SF", "Year.Built")])

# Data Preparation: Impute Missing Values

cat("\n--- Data Preparation ---\n")
# Identify numeric columns
numeric_vars <- sapply(ames_data, is.numeric)
numeric_data <- ames_data[, numeric_vars]
```

```
# Impute missing values with the mean for each numeric column
for(col in 1:ncol(numeric_data)) {
  if(any(is.na(numeric_data[,col]))) {
    mean_val <- mean(numeric_data[,col], na.rm = TRUE)
    numeric_data[is.na(numeric_data[,col]), col] <- mean_val
    cat("Missing values in ", colnames(numeric_data)[col], " imputed with mean: ",
round(mean_val, 2), "\n")
  }
}

# Verify that there are no more NA values in the numeric subset
cat("\nTotal NA values in numeric data after imputation: ", sum(is.na(numeric_data)),
"\n")

# Correlation Matrix

cat("\n--- Correlation Analysis ---\n")
# Calculate the correlation matrix
cor_matrix <- cor(numeric_data)
# View the correlations with SalePrice
sale_price_cor <- cor_matrix["SalePrice", ]
cat("\nTop 10 correlations with SalePrice:\n")
print(head(sort(abs(sale_price_cor), decreasing = TRUE), 11))

# Plot the Correlation Matrix

# Set graphical parameters for better visualization
par(mfrow=c(1,1))
# Generate the correlation plot
corrplot(cor_matrix, method = "color", type = "upper", order = "hclust",
  tl.col = "black", tl.srt = 45, tl.cex = 0.6,
  title = "Correlation Matrix of Ames Housing Numeric Variables",
  mar=c(0,0,1,0))

# Scatter Plots

# Find variable with the highest correlation with SalePrice (excluding SalePrice itself)
highest_cor_var <- names(sort(abs(sale_price_cor), decreasing = TRUE)[2])
highest_cor_val <- sale_price_cor[highest_cor_var]

# Find variable with the lowest absolute correlation with SalePrice
# We exclude PID and Order as they are identifiers
lowest_cor_var <- names(sort(abs(sale_price_cor[!names(sale_price_cor) %in%
c("PID", "Order") ]), decreasing = FALSE)[1])
```

```
lowest_cor_val <- sale_price_cor[lowest_cor_var]

# Find variable with correlation closest to 0.5
mid_cor_var <- names(which.min(abs(sale_price_cor - 0.5)))
mid_cor_val <- sale_price_cor[mid_cor_var]

cat("\nHighest correlation with SalePrice:", highest_cor_var, "(",
round(highest_cor_val, 3), ")\n")
cat("Lowest absolute correlation with SalePrice:", lowest_cor_var, "(",
round(lowest_cor_val, 3), ")\n")
cat("Correlation closest to 0.5 with SalePrice:", mid_cor_var, "(", round(mid_cor_val,
3), ")\n")

# Create a 1x3 layout for plots
par(mfrow=c(1,3), mar=c(4, 4, 2, 1))

# Scatter plot for the highest correlation
plot(numeric_data[[highest_cor_var]], numeric_data$SalePrice,
     main = paste("Highest Correlation:\n", highest_cor_var),
     xlab = highest_cor_var, ylab = "Sale Price", pch = 16, col = "blue")

# Scatter plot for the lowest correlation
plot(numeric_data[[lowest_cor_var]], numeric_data$SalePrice,
     main = paste("Lowest Correlation:\n", lowest_cor_var),
     xlab = lowest_cor_var, ylab = "Sale Price", pch = 16, col = "grey")

# Scatter plot for correlation closest to 0.5
plot(numeric_data[[mid_cor_var]], numeric_data$SalePrice,
     main = paste("Mid Correlation (~0.5):\n", mid_cor_var),
     xlab = mid_cor_var, ylab = "Sale Price", pch = 16, col = "orange")

# Reset plotting layout
par(mfrow=c(1,1))

#Fit an Initial Regression Model

cat("\n--- Initial Regression Model ---\n")
# Select three continuous variables based on high correlation and domain knowledge
# Let's use Overall.Qual, Gr.Liv.Area, and Total.Bsmt.SF
initial_model <- lm(SalePrice ~ Overall.Qual + Gr.Liv.Area + Total.Bsmt.SF, data =
numeric_data)

# Print model summary
cat("\nSummary of Initial Model:\n")
summary(initial_model)
```

```
# Model Equation and Coefficient Interpretation
```

```
cat("\nModel Equation: SalePrice = ", round(coef(initial_model)[1], 2),  
    " + ", round(coef(initial_model)[2], 2), " * Overall.Qual",  
    " + ", round(coef(initial_model)[3], 2), " * Gr.Liv.Area",  
    " + ", round(coef(initial_model)[4], 2), " * Total.Bsmt.SF\n")
```

```
# Plot and Interpret Diagnostic Graphs
```

```
cat("\n--- Model Diagnostics ---\n")  
# Plotting the diagnostic graphs in a 2x2 layout  
par(mfrow=c(2,2))  
plot(initial_model)  
# Reset plotting layout  
par(mfrow=c(1,1))
```

```
# Check for Multicollinearity
```

```
cat("\nChecking for Multicollinearity using VIF:\n")  
# Calculate Variance Inflation Factor (VIF)  
vif_values <- vif(initial_model)  
print(vif_values)
```

```
# Check for Outliers/Influential Points
```

```
cat("\nChecking for Outliers and Influential Points:\n")  
# Outliers can be seen in the Residuals vs. Fitted plot  
# Influential points can be identified using Cook's distance  
# The diagnostic plots from step 9 show points 1499 and 2181 as potential issues.  
cat("Points 1499, 2181, 2182 appear to be influential based on diagnostic plots.\n")  
# Let's inspect these observations  
influential_points <- numeric_data[c(1499, 2181, 2182), c("SalePrice", "Overall.Qual",  
    "Gr.Liv.Area", "Total.Bsmt.SF")]  
cat("\nData for influential points:\n")  
print(influential_points)
```

```
# Attempt to Correct Model Issues
```

```
cat("\n--- Model Improvement ---\n")  
# The diagnostic plots suggest non-normality of residuals and heteroscedasticity.  
# A log transformation of the response variable (SalePrice) often helps.  
# Let's also remove the two most extreme outliers identified (2181, 2182) which have  
# huge Gr.Liv.Area.  
numeric_data_cleaned <- numeric_data[-c(2181, 2182), ]
```

```
# Fit a new model with log-transformed SalePrice and cleaned data
improved_model <- lm(log(SalePrice) ~ Overall.Qual + Gr.Liv.Area + Total.Bsmt.SF,
data = numeric_data_cleaned)

cat("\nSummary of Improved Model (log-transformed SalePrice, outliers
removed):\n")
summary(improved_model)

# Plot diagnostics for the improved model
par(mfrow=c(2,2))
plot(improved_model)
par(mfrow=c(1,1))

# All Subsets Regression to Find the "Best" Model

cat("\n--- All Subsets Regression ---\n")
# We need a dataframe without the response variable for the predictor set
# Let's select a reasonable number of top predictors to avoid excessive computation
top_predictors_names <- names(head(sort(abs(sale_price_cor), decreasing = TRUE),
15))
predictors_df <- numeric_data_cleaned[, top_predictors_names]
predictors_df$SalePrice <- NULL # Remove response variable

# Perform all subsets regression
# Note: This can be computationally intensive
subsets <- regsubsets(log(SalePrice) ~ ., data = cbind(predictors_df, SalePrice =
numeric_data_cleaned$SalePrice), nvmax = 14)
subsets_summary <- summary(subsets)

# Plot results to select the best model based on a criterion (e.g., Adjusted R^2)
par(mfrow=c(1,2))
plot(subsets_summary$adjr2, xlab="Number of Variables", ylab="Adjusted R-
squared", type="b")
plot(subsets_summary$bic, xlab="Number of Variables", ylab="BIC", type="b")
par(mfrow=c(1,1))

# Find the model with the highest Adjusted R-squared
best_model_size <- which.max(subsets_summary$adjr2)
cat("\nBest model size based on Adjusted R-squared:", best_model_size,
"variables.\n")

# Get the coefficients/variables for the best model
best_model_vars <- names(coef(subsets, id=best_model_size))
# Create the formula for the best model
```

```
best_model_formula <- as.formula(paste("log(SalePrice) ~", paste(best_model_vars[-
1], collapse="+")))

# Fit the final "best" model
best_model <- lm(best_model_formula, data = numeric_data_cleaned)

cat("\nSummary of the 'Best' Model from all subsets regression:\n")
summary(best_model)

# 14. Compare Models
cat("\n--- Final Model Comparison ---\n")
cat("\n--- Improved Manual Model (Step 12) ---\n")
print(summary(improved_model))
cat("\n--- 'Best' All Subsets Model (Step 13) ---\n")
print(summary(best_model))

cat("\nThe 'Best' model from all subsets regression includes more variables and has a
higher Adjusted R-squared (",
    round(summary(best_model)$adj.r.squared, 4), ") compared to the improved 3-
variable model (",
    round(summary(improved_model)$adj.r.squared, 4), ").\n")

# Report the best model equation
best_coefs <- coef(best_model)
cat("\nEquation for the 'Best' Model:\n log(SalePrice) = ", round(best_coefs[1], 2))
for (i in 2:length(best_coefs)) {
  cat(" + (", round(best_coefs[i], 4), " * ", names(best_coefs)[i], ")")
}
cat("\n")

#End
```